BanglaTense: A Large-Scale Dataset of Bangla Sentences Categorized by Tense: Past, Present, and Future

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Data Article

BanglaTense: A large-scale dataset of Bangla sentences categorized by tense: Past, present, and future

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A B S T R A C T

Bengali, an Indo-Aryan language, features a complex grammatical structure with tenses, which is crucial for natural language processing (NLP) applications like text classification, machine translation, and sentiment analysis. The BanglaTense dataset is a large-scale, meticulously curated collection of Bangla sentences categorized by their tense: Past, present, and future. Addressing the resource gap in NLP for the Bangla language, BanglaTense provides a curated resource for Bangla sentence classification, featuring 17,819 annotated sentences, with 5,629 in the past tense, 6,101 in the present tense, and 6,089 in the future tense. This dataset is a benchmark for evaluating NLP models on Bangla sentence classification, promoting linguistic diversity and inclusive language models while ensuring balanced representation across categories. Preprocessing steps are applied to enhance data quality, including anonymization and duplicate removal. Three native Bangla speakers independently assessed the tense labels of the sentences, ensuring the dataset's reliability. BanglaTense is designed to advance research and development in NLP for Bangla, offering valuable applications in tense detection, text classification, language modeling, and educational tools. This dataset supports linguistic study and enhances the development of precise and context-aware NLP models by providing a robust foundation for temporal analysis in Bangla sentences. The dataset is openly available for academic and

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research purposes, promoting collaboration and innovation within the Bangla NLP community.

Specifications Table

Subject	Computer Science
Specific subject area	Machine Learning, Natural Language Processing, Bangla Text Classification, Bangla Tense Classification
Type of data	Text Files (xlsx-formatted)
Data collection	The data for the BanglaTense dataset was extracted from various Bangla sources, including blogs, Facebook pages, literature, daily conversations, educational content, and news articles from Bangladesh. This selective collection process was designed to ensure a balanced data distribution across each past, present, and future tense category. The dataset comprises 17,819 sentences, with 5,629 sentences in the past tense, 6,101 in the present tense, and 6,089 in the future tense. Native Bangla speakers meticulously annotated the data to guarantee high accuracy and consistency, ensuring the reliability of the dataset for natural language processing applications.
Data source location	Publicly open Bangla blogs, Facebook pages, magazines and news articles
Data accessibility	Repository name: Mendeley Data Data identification number: 10.17632/39w5khrg87.4 Direct URL to data: https://data.mendeley.com/datasets/39w5khrg87/4

1. Value of the Data

- The BanglaTense dataset provides a robust foundation for developing and training models for tense detection in Bangla sentences. This is crucial for various NLP applications, including text summarization, machine translation, and dialogue systems, where understanding the temporal context is essential.
- By categorizing sentences into past, present, and future tenses, the dataset aids in the creation of classifiers that can accurately determine the tense of a given sentence. This enhances the performance of text classification systems, making them more precise in tasks that require temporal understanding.
- The dataset improves language models by providing the necessary data to learn the nuances of tense usage in Bangla. This results in more accurate and context-aware language models, which are beneficial for applications such as predictive text input and automated content generation.
- For linguists and language researchers, BanglaTense offers a rich resource for studying the grammatical structure and temporal aspects of Bangla. It allows for in-depth analysis and comparison of tense usage patterns, contributing to the broader field of computational linguistics.
- The dataset is a valuable educational resource for teaching and learning Bangla grammar, particularly in understanding and practicing the correct use of tenses. It can be integrated into language learning applications and educational platforms to provide learners with practical examples and exercises.
- By setting a benchmark for tense categorization in Bangla, Bangladesh encourages the development of similar datasets for other languages, fostering a more inclusive and comprehensive approach to NLP research across different linguistic contexts.

2. Background

The original motivation behind compiling the BanglaTense dataset stems from the need to address the resource gap in natural language processing (NLP) for the Bangla language, which is spoken by millions of people but lacks comprehensive datasets for advanced linguistic analysis [1]. Theoretical underpinnings highlight the importance of tense in understanding temporal context, which is crucial for NLP applications such as text classification, machine translation, and sentiment analysis. Methodologically, the dataset aims to provide a balanced and diverse representation of Bangla sentences across different tenses, enabling the development of more accurate and context-aware NLP models. By categorizing sentences into past, present, and future tenses, the dataset supports creating classifiers and language models that effectively handle temporal aspects of language. The meticulous annotation by native Bangla speakers ensures high accuracy and consistency, making the dataset a reliable resource for researchers and developers [2]. The BanglaTense dataset fills a significant gap in Bangla NLP resources and serves as a benchmark for evaluating and improving NLP models in the context of tense classification.

3. Data Description

Bengali, one of the most prominent Eastern Indo-Aryan languages in the Indian subcontinent, ranks as the 7th most spoken language globally, with approximately 272.8 million native speakers [3]. This language, known for its intricate grammatical structure and tenses, poses significant challenges for natural language processing (NLP) applications such as text classification, machine translation, and sentiment analysis. Bengali text's complex morphology and syntax make automating tense identification and classification difficult. To address this challenge, we have developed the “BanglaTense” dataset, a meticulously curated collection of 17,819 sentences designed explicitly for this study. This dataset is available as the raw data file “Cleaned-BanglaTense.xlsx” in the repository.

The BanglaTense dataset has been sourced from various publicly accessible Bangla blogs, Facebook pages, magazines, and news articles, ensuring a diverse representation of contemporary language use. A critical aspect of the dataset's curation was maintaining an equal distribution of sentences across three tense categories: past, present, and future. This balanced representation is crucial for training models that can generalize effectively across linguistic structures and complexities. Bengali belongs to the Indo-Aryan branch of the Indo-European language family and is closely related to Assamese and Odia. It is the primary language of Bangladesh and the Indian states of West Bengal, Tripura, and Assam, and it is also spoken by diaspora communities worldwide. As the official language of Bangladesh and one of the 22nd scheduled languages of India, Bengali holds significant cultural and linguistic importance, with a global speaker population estimated at approximately 272.8 million [4].

The following description provides a detailed overview of the variables in the BanglaTense dataset, enabling researchers to fully utilize this resource to advance NLP applications in the Bengali language. As outlined in Table 1, the dataset structure details the various components

Table 1
Dataset description with attributes and possible values.

Attribute	Description and Possible values
Sentence	This attribute represents the original text in Bengali. Each Bangla sentence in the dataset was obtained from publicly accessible sources, including Bangla blogs, Facebook pages, magazines, and news articles. Example: পরিস্থিতি এই মুখের হাসি কেড়ে নিয়েছে (Circumstances have robbed this face of a smile) এই ছবির রেকর্ড এখনও কেউ ভাঙতে পারেনি (No one has broken the record of this film yet) আমরা আগামীকাল একটি নতুন সিনেমা দেখবো (We will watch a new movie tomorrow)
Tense	This attribute indicates the tense category of the text, classifying each sentence as either Past, Present, or Future tense. Example: অতীত কাল (Past tense) বর্তমান কাল (Present tense) ভবিষ্যৎ কাল (Future tense)

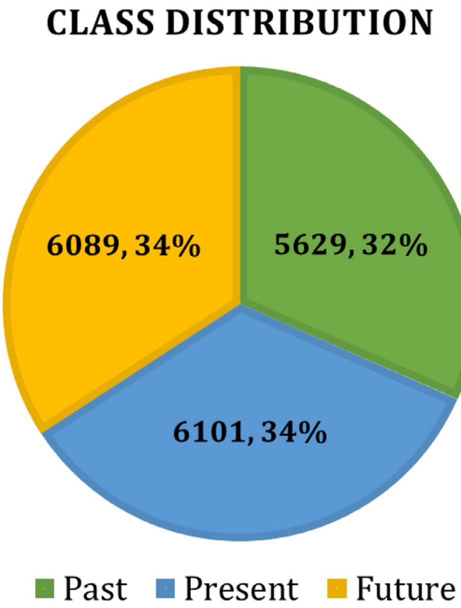

Fig. 1. The class distribution of each category of tense.

and their respective annotations, facilitating its use for research and development purposes. Fig. 1 illustrates the class distribution across the three categories of tense within the dataset: 5,629 sentences (32%) in the past tense, 6,101 sentences (34%) in the present tense, and 6,089 sentences (34%) in the future tense.

Fig. 2 demonstrates the distribution of text length within the BanglaTense dataset, which classifies sentences into past, present, and future tenses. For past tense sentences, there are 5,629 instances with an average length of 35.02 characters and a standard deviation of 23.59 characters. These sentences range from 9 to 263 characters, with the middle 50% falling between 23 and 29 characters and a median length of 36 characters. The present tense category includes 6,101 sentences, averaging 36.50 characters in length with a standard deviation of 22.68 characters. Sentence lengths vary from 9 to 203 characters, with the middle 50% ranging from 22 to 40 characters and a median length of 29 characters. Future tense sentences comprise 6,089 instances, averaging 37.96 characters with a standard deviation of 26.18 characters. These sentences range from 10 to 217 characters, with the middle 50% between 21 and 43 characters and a median length of 29 characters.

Overall, the BanglaTense dataset shows a balanced distribution of sentences across the three tenses, with past and future tense sentences being slightly longer on average compared to present tense sentences. The variability in sentence lengths is similar across categories, with the present tense showing the longest maximum sentence length. This comprehensive statistical overview helps in understanding the characteristics of the dataset, which is crucial for developing and validating models for NLP applications in Bengali.

Table 2 presents the 20 most frequent words in past, present, and future tense sentences from the BanglaTense dataset, along with their frequencies. “ছিল” (was) appears most frequently in past tense sentences (867 times), highlighting its common use for describing past events. In present tense, “করে” (by doing) stands out (284 times), commonly used to indicate ongoing actions. For future tense, “হবে” (wil be) leads with 1,513 occurrences, frequently used to express future plans or scenarios.

However, the analysis includes stopwords, common words with minimal meaning typically excluded in text analysis. Their inclusion may not offer a fully comprehensive view of the

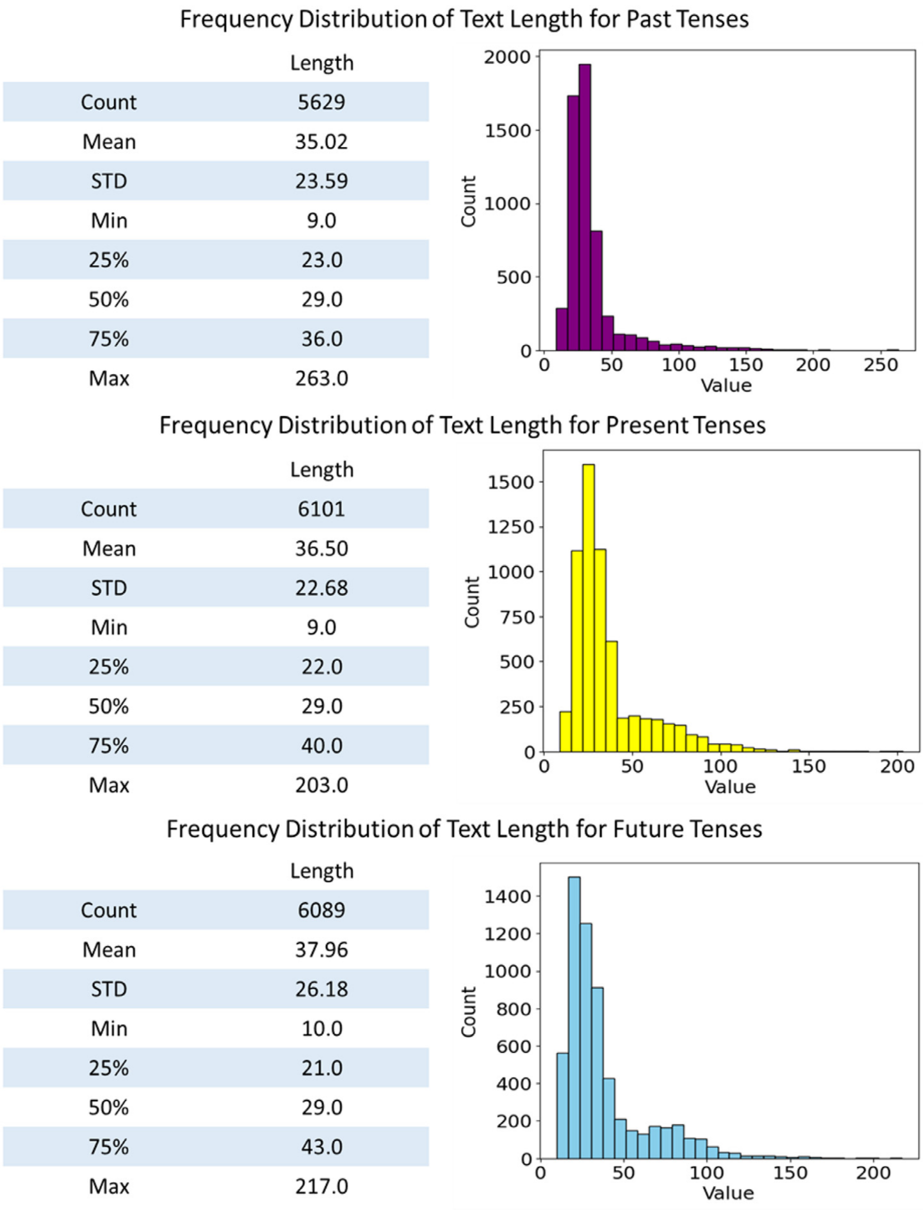

Fig. 2. The frequency distribution of Bangla text length for past, present and future tenses.

dataset's most significant terms [5]. Despite this limitation, examining high-frequency words provides valuable insights into prevalent vocabulary and linguistic patterns within the dataset. This understanding is crucial for developing effective language models and algorithms, enhancing accuracy in predictions and classifications. Additionally, focusing on these words supports important data preprocessing tasks such as stop-word removal and feature selection, contributing to refining the dataset for training Natural Language Processing (NLP) models.

Table 2
Most common 20 words of past, presents and futures tenses with its frequency.

Past Tense			Present Tense			Future Tense		
Bangla Word	English Form	Frequency	Bangla Word	English Form	Frequency	Bangla Word	English Form	Frequency
ছিল	Was	867	করে	By doing	284	হবে	Will be	1513
হয়েছিল	Was done	226	করি	I do	277	করবে	Will do	665
করছিলাম	Was doing	217	করুন	Do	251	একটি	A	593
করলাম	I did	185	হচ্ছে	Being	234	করব	Will do	548
করেছিলাম	I did	153	করছি	Doing	210	থাকবে	Will be	531
করেছিল	Did	151	হয়	Is	185	থাকব	Will stay	462
হচ্ছিল	Was happening	132	করা	To do	139	যাবে	Will go	189
দেখেছিলাম	I saw	128	যায়	Goes	122	পাবে	Will get	115
হল	Hall	80	চাই	Want	98	যাব	Will go	103
করতাম	I did	79	করেছি	I did	88	আসবে	Will come	85
করছিল	Was doing	74	হয়ে	Become	87	করিব	Will do	71
করল	Did	74	করছে	Doing	83	যাবো	I will go	70
লেগেছিল	It took	70	পারে	Can	79	করবেন	Will do	69
করেছিলেন	Did	69	থাকে	Remains	79	হব	Will be	68
করতেছিলাম	Was doing	66	হয়েছে	Has been	72	থাকবো	Will stay	64
চাইলাম	I wanted	66	যাচ্ছে	Is going	71	চাইব	I want	64
করিয়াছিলাম	I did	63	করিয়াছি	Have done	70	দিবে	Will give	62
চাচ্ছিলাম	I wanted	60	গেছে	Gone	64	লাগবে	Will need	59
দেখলাম	I saw	56	করতেছি	Doing	62	চলবে	Will continue	53
ছিলেন	Was	56	চাচ্ছি	I want	61	পাবো	Will get	49

Fig. 3. Bangla word cloud for past, present and future tenses within BanglaTense dataset.

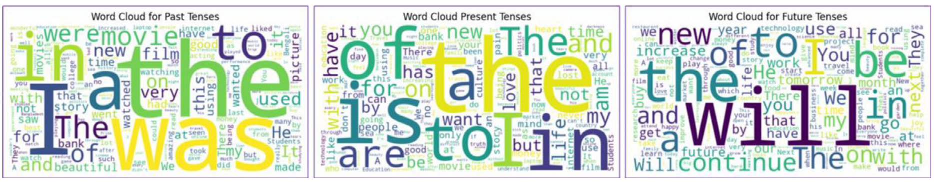

Fig. 4. Corresponding word cloud for past, present and future tenses within BanglaTense dataset in English form.

Fig. 3 showcases word clouds representing the past, present, and future tense categories of the BanglaTense dataset. Fig. 4 presents the corresponding word clouds in the English language. From the past tense word cloud, it is clearly observed that the three most common words are “ছলি” (was), “হয়ছেলি” (was done), and “করছলিাম” (was doing). In the present tense word cloud, the most frequent words are “আমি” (I), “করি” (I do), and “করুন” (do). For the future tense, the most prominent words are “হবে” (will be), “করবে” (will do), and “করব” (will do). Each word cloud visually depicts the frequency and prominence of words within its respective tense category based on their occurrence in the dataset, providing a clear and intuitive representation of language usage trends across different temporal contexts.

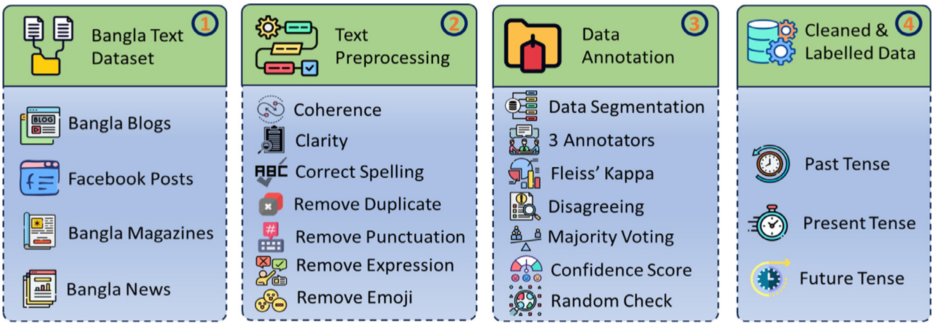

Fig. 5. Data collection and preprocessing to create BanglaTense dataset.

4. Experimental Design, Materials and Methods

To generate a substantial volume of text data, particularly in Bangla, which poses significant challenges, the BanglaTense dataset creation process adheres to a structured approach. Initially, Bangla text was sourced from publicly available sources such as blogs, Facebook pages, magazines, and news articles. The content was compiled into a Google Spreadsheet file for aggregation. Subsequently, the dataset underwent rigorous preprocessing steps, including initial cleaning, anonymization, removal of duplicates, and filtering out any instances of profane language.

In the next stage, the dataset underwent meticulous evaluation by three native Bangla speakers. Each evaluator independently labeled sentences according to three distinct tense categories: past, present, and future. This thorough assessment ensured accurate categorization of sentences based on tense, enhancing the reliability and quality of the dataset. Fig. 5 illustrates the systematic workflow of the dataset generation process, depicting each step from data collection to final labeling by native speakers.

Following the collection of raw Bangla text, the processing of the BanglaTense dataset involves several critical steps to ensure its quality and usability. The process begins with primary cleaning to enhance coherence, clarity, and correctness, including rectifying or removing spelling errors and other inconsistencies manually to maintain data integrity. Duplicate entries are identified and removed to avoid redundancy, ensuring the uniqueness of each data point. Punctuation marks, which typically do not contribute to linguistic analysis, are eliminated, and regular expressions are applied to detect and remove unwanted patterns. Emojis, which are generally irrelevant for text-based language modeling, are also removed. These comprehensive preprocessing steps collectively result in a dataset that is clean, accurate, and ready for subsequent analysis and modeling tasks.

During the data annotation phase, the Inter-Annotator Agreement (IAA) [6] was evaluated using the Fleiss Kappa coefficient, as detailed in Algorithm 1. The Fleiss Kappa coefficient is crucial for quantifying the consistency of annotators when categorizing text into past, present, and future tenses. A high Fleiss Kappa value reflects strong agreement among annotators, ensuring the reliability of the labeled data. This metric also plays a pivotal role in validating the dataset's robustness, as it helps identify and resolve inconsistencies during the annotation process. Finally, we obtain the cleaned and labeled data categorized as past, present, and future tense, respectively.

After the completion of data annotation, four deep learning techniques were evaluated to classify Bangla tense into three categories: past, present, and future. The model training was conducted with a batch size of 64 over 50 epochs. After implementing the models, five performance metrics were computed to identify the best-performing model for tense classification. Among the evaluated models, the BiLSTM demonstrated the highest average accuracy of 95.92%, outperforming the other approaches. Conversely, the LSTM model achieved the lowest average

Algorithm 1 BanglaTense data annotation process.

Input:	Raw Bangla text, B_{Text}
Output:	Tense of Bangla text, B_{Tense}
Step-1:	Segment the sentences into manageable batches for annotation (i.e., 6000 sentences per speakers)
Step-2:	Select a group of 3 native Bengali speakers with strong language proficiency
Step-3:	Assign each batch of sentences to multiple annotators (e.g., 3 annotators per sentence) to ensure redundancy
Step-4:	Use 'Google Spreadsheet' where annotators can label each sentence as past, present, or future tense
Step-5:	Calculate agreement using statistical measures (i.e., Fleiss' Kappa) to calculate the inter-annotator agreement (IAA)
Step-6:	Calculate Fleiss' Kappa:
Step-7:	$\mathcal{K} = \frac{\bar{P}-\bar{P}_e}{1-\bar{P}_e} \begin{cases} Where,\ \bar{P}\ is\ the\ mean\ of\ the\ observed\ agreement, \\ \bar{P}_e\ is\ the\ mean\ of\ the\ expected\ agreement\ by\ chance \end{cases}$
Step-8:	Checking conflict of resolution by reviewing disagreement and sort them for further review and annotate with expert annotator
Step-9:	Measuring majority voting by counting the number of votes for each tense and set the final annotation with the highest vote
Step-10:	Calculate the confidence score for each sentence based on the proportion of annotators who agreed on the label
Step-11:	$Confidence\ Score = \frac{Number\ of\ AnnotatorsAgreeing}{Total\ Number\ of\ Annotators} \{\ A\ high\ confidence\ score\ indicates\ the\ high\ reliablity$
Step-12:	Checking randomly samples annotated sentences and perform a quality check to ensure annotations meet the required standards
Step-13:	Compile the final annotated sentences into the BanglaTense dataset

Table 3
Performance of applied four neural network-based deep learning models on BanglaTense dataset.

Model	Class	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC
LSTM	Past	93.90	91.12	91.02	91.07	0.99
	Present	94.69	91.93	91.28	91.61	0.99
	Future	97.94	96.62	97.37	96.99	1.00
BiLSTM	Past	94.65	92.87	91.35	92.10	0.99
	Present	95.32	92.79	92.46	92.63	0.99
	Future	97.79	96.00	97.59	96.79	0.99
Conv1D-LSTM	Past	94.01	94.24	87.84	90.93	0.99
	Present	94.31	87.84	95.29	91.41	0.99
	Future	98.50	98.34	97.26	97.79	0.99
Conv1d-BiLSTM	Past	94.24	92.78	90.14	91.44	0.99
	Present	94.61	89.65	93.88	91.71	0.99
	Future	98.43	98.44	96.93	97.68	0.99

accuracy at 95.51%. For class-wise performance, the Conv1D-LSTM model excelled in classifying the Future tense, achieving an impressive accuracy of 98.50%.

In addition to accuracy, the BiLSTM model performed exceptionally well on other evaluation metrics, including precision (93.89%), recall (93.80%), F1 score (93.84%), and an AUC value of 0.99, indicating near-perfect performance. These results, along with detailed comparisons, are summarized in Table 3.

Limitations

The BanglaTense dataset, though valuable for Bengali NLP, has several limitations. Including stopwords may reduce the informativeness of word frequency analysis. Manual labeling by three native speakers introduces potential bias and inconsistency. The dataset's sources—blogs, Facebook pages, magazines, and news articles—might not fully represent the diversity of Bangla language usage, affecting generalizability. Preprocessing steps could inadvertently remove relevant information along with noise. Additionally, focusing only on past, present, and future tenses might oversimplify the complexity of Bengali verb tenses, overlooking nuanced grammatical structures. Despite these issues, the BanglaTense dataset is a significant resource for Bengali NLP research.

Ethics Statement

The BanglaTense dataset prioritizes ethical data collection practices. Publicly available content from blogs, Facebook pages, magazines, and news articles was responsibly sourced and incorporated. Strict criteria ensured inclusion of only copyright-free material, safeguarding against potential infringement. Furthermore, the dataset upholds responsible use principles, preventing any harm or violation of individual rights. For content retrieved from Facebook pages, adherence to their content usage policies guaranteed no need for additional permissions [7].

CRediT Author Statement

Md. Hasan Imam Bijoy: Conceptualization, Methodology, Visualization, Software, Resources, Supervision, Writing, Original draft preparation, Writing –review & editing; **Umme Ayman:** Data curation, Methodology, Visualization, Investigation, Validation, Writing –review & editing; **Md. Monarul Islam:** Data curation, Investigation, Validation, Writing –review & editing.

Data availability

BanglaTense: A Large-Scale Dataset of Bangla Sentences Categorized by Tense: Past, Present, and Future (Original data) (Mendeley Data).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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